

Set

Neeraj Varghese

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Mathematical Form (1–5)

1. $\{1, 2, 3, 4, 5, 6\}$

$$X = \{x : x \in \mathbb{N}, 1 \leq x \leq 6\}$$

2. $\{2, 4, 6, 8, 10\}$

$$X = \{x : x \in \mathbb{N}, x \equiv 0 \pmod{2}, 2 \leq x \leq 10\}$$

3. $\{1, 3, 5, 7, 9\}$

$$X = \{x : x \in \mathbb{N}, x \equiv 1 \pmod{2}, 1 \leq x \leq 9\}$$

4. $\{5, 10, 15, 20, 25\}$

$$X = \{x : x = 5n, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

5. $\{1, 4, 9, 16, 25\}$

$$X = \{x : x = n^2, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

Mathematical Form (6–10)

6. $\{1, 8, 27, 64, 125\}$

$$X = \{x : x = n^3, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

7. $\{2, 3, 5, 7, 11\}$

$$X = \{x : x \text{ is prime}, 2 \leq x \leq 11\}$$

8. $\{10, 20, 30, 40, 50\}$

$$X = \{x : x = 10n, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

9. $\{3, 6, 9, 12, 15\}$

$$X = \{x : x = 3n, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

10. $\{0, 1, 2, 3, 4, 5\}$

$$X = \{x : x \in \mathbb{Z}, 0 \leq x \leq 5\}$$

General Inclusion–Exclusion Principle

For finite sets $A_1, A_2, \dots, A_n$,

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= \sum_{i=1}^n |A_i| \\ &\quad - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ &\quad + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| \\ &\quad - \dots \\ &\quad + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned}$$

Cardinality of a Set & Finite Set

Cardinality of a Set

The cardinality of a set is the number of distinct elements in the set. It is denoted by $|A|$.

Example

$$A = \{\text{Python}, \text{R}, \text{SQL}, \text{TensorFlow}\}$$

$$|A| = 4$$

Finite Set

A finite set contains a limited number of elements.

$$B = \{\text{CNN}, \text{RNN}, \text{Transformer}, \text{GAN}\}$$

Since

$$|B| = 4,$$

B is a finite set.

Infinite Set & Null Set

Infinite Set

A set containing infinitely many elements.

Example

All possible values of neural network weights

$$W = \{w : w \in \mathbb{R}\}$$

Hence,

$$|W| = \infty$$

Null (Empty) Set

A set containing no elements.

Notation:

$$\emptyset$$

Example:

Suppose no student completed an AI Ethics certification.

Equal Set

Definition

Two sets are equal if they contain exactly the same elements.

Example

$$A = \{\text{Python}, \text{SQL}, \text{Java}\}$$

$$B = \{\text{Java}, \text{Python}, \text{SQL}\}$$

$$A = B$$

Equivalence Relation

Definition

A relation R on a set S is an equivalence relation if it is:

Reflexive, Symmetric, Transitive

Example

$$S = \{\text{CNN}, \text{ResNet}, \text{LSTM}, \text{GRU}\}$$

Define

$$xRy \iff x \text{ and } y \text{ belong to the same model family.}$$

Examples:

$$\text{CNN } R \text{ ResNet}, \quad \text{LSTM } R \text{ GRU}$$

Hence, R is an equivalence relation.

Subset

Definition

A set A is called a subset of B if every element of A is also an element of B .

Notation:

$$A \subseteq B$$

Example

$$A = \{\text{Python}, \text{SQL}\}$$

$$B = \{\text{Python}, \text{SQL}, \text{Java}, \text{R}\}$$

$$A \subseteq B$$

Universal Set

Definition

The universal set contains all objects under consideration.

Notation:

$$U$$

Example

$$U = \{\text{Python, Java, C++}, \text{SQL}, \text{R}\}$$

Superset

Definition

A set B is called a superset of A if every element of A belongs to B .

Notation:

$$B \supseteq A$$

Example

$$A = \{\text{Python}, \text{SQL}\}$$

$$B = \{\text{Python}, \text{SQL}, \text{Java}\}$$

$$B \supseteq A$$

Power Set

Definition

The power set is the set of all subsets of a set.

Notation:

$$P(A)$$

If

$$|A| = n,$$

then

$$|P(A)| = 2^n$$

Example

$$A = \{\text{Python}, \text{SQL}\}$$

$$P(A) = \{\emptyset, \{\text{Python}\}, \{\text{SQL}\}, \{\text{Python}, \text{SQL}\}\}$$

Singleton Set

Definition

A singleton set contains exactly one element.

Example

$$A = \{\text{TensorFlow}\}$$

Since

$$|A| = 1,$$

A is a singleton set.

Union

Definition

The union of two sets contains all elements that belong to either set.

Notation:

$$A \cup B$$

Example

$$A = \{\text{Python}, \text{R}\}$$

$$B = \{\text{SQL}, \text{Python}\}$$

$$A \cup B = \{\text{Python}, \text{R}, \text{SQL}\}$$

Intersection

Definition

The intersection of two sets contains the common elements of both sets.

Notation:

$$A \cap B$$

Example

$$A = \{\text{Python}, \text{SQL}, \text{Java}\}$$

$$B = \{\text{Python}, \text{R}, \text{SQL}\}$$

$$A \cap B = \{\text{Python}, \text{SQL}\}$$

Set Difference

Definition

The set difference contains elements that are in A but not in B .

Notation:

$$A - B$$

Example

$$A = \{\text{Python, SQL, Java}\}$$

$$B = \{\text{Python, R}\}$$

$$A - B = \{\text{SQL, Java}\}$$

Symmetric Difference

Definition

The symmetric difference contains elements that belong to exactly one of the two sets.

Notation:

$$A \triangle B$$

Formula:

$$A \triangle B = (A - B) \cup (B - A)$$

Example

$$A = \{\text{Python}, \text{SQL}\}$$

$$B = \{\text{Python}, \text{R}\}$$

$$A \triangle B = \{\text{SQL}, \text{R}\}$$

Complement

Definition

The complement of a set contains all elements in the universal set that are not in the set.

Notation:

$$A^c$$

Example

$$U = \{\text{Python, Java, SQL, R}\}$$

$$A = \{\text{Python, SQL}\}$$

$$A^c = \{\text{Java, R}\}$$

Cartesian Product

Definition

The Cartesian product is the set of all ordered pairs formed by taking one element from each set.

Notation:

$$A \times B$$

Example

$$A = \{\text{CNN}, \text{RNN}\}$$

$$B = \{\text{TensorFlow}, \text{PyTorch}\}$$

$$A \times B = \left\{ \begin{array}{l} (\text{CNN}, \text{TensorFlow}), \\ (\text{CNN}, \text{PyTorch}), \\ (\text{RNN}, \text{TensorFlow}), \\ (\text{RNN}, \text{PyTorch}) \end{array} \right\}$$

Partially Ordered Set (Poset)

Definition

A relation R on a set A is called a **partially ordered relation** if it satisfies:

- ▶ Reflexive: $aRa, \forall a \in A$
- ▶ Antisymmetric: If aRb and bRa , then $a = b$
- ▶ Transitive: If aRb and bRc , then aRc

The pair (A, R) is called a **Partially Ordered Set (Poset)**.

Example:

The relation "divides" ($|$) on the set

$$A = \{1, 2, 3, 6\}$$

is a partially ordered relation.

Example 1: Divisibility Relation

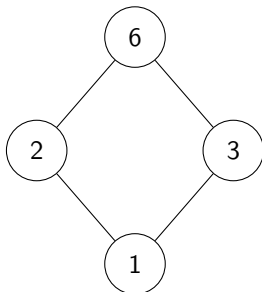
Let

$$A = \{1, 2, 3, 6\}$$

Define

$$aRb \iff a \mid b$$

where " $\mid$ " means "divides".



Hasse Diagram: The diagram shows the divisibility order after removing reflexive and transitive edges.

Example 2: Subset Relation

Let

$$A = \{a, b\}$$

The power set of A is

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

Define the relation

$$XRY \iff X \subseteq Y$$

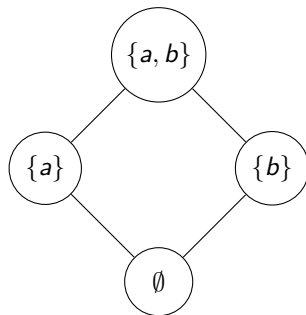
where " $\subseteq$ " means "is a subset of".

Since the subset relation is reflexive, antisymmetric and transitive,

$$(P(A), \subseteq)$$

is a partially ordered set.

Hasse Diagram for $(P(A), \subseteq)$



Observation

The Hasse diagram represents the subset ordering of the power set. Only the covering relations are shown.

Example 3: AI Model Hierarchy

Let

$$A = \{\text{AI, Machine Learning, Deep Learning, CNN, RNN}\}$$

Define the relation

$$xRy \iff x \text{ is a subtype of } y.$$

Examples:

CNN R Deep Learning

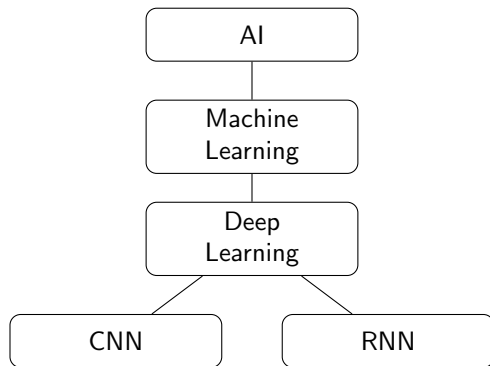
RNN R Deep Learning

Deep Learning R Machine Learning

Machine Learning R AI

Hence, (A, R) is a partially ordered set.

Hasse Diagram for AI Model Hierarchy



Observation

The diagram shows the hierarchy of AI models. CNN and RNN are incomparable, but both are covered by Deep Learning.